Gemcitabine (1250mg/m^2) Monotherapy - 21 day

INDICATIONS FOR USE:

INDICATION	ICD10	Regimen Code	HSE approved reimbursement status *
Loco-regionally recurrent/metastatic nasopharyngeal cancer not amenable for local curative therapy [i]	C11	00514a	N/A

* This applies to post 2012 indications

TREATMENT:

The starting dose of the drugs detailed below may be adjusted downward by the prescribing clinician, using their independent medical judgement, to consider each patients individual clinical circumstances.

Gemcitabine is administered once weekly for 2 consecutive weeks followed by a 1 week pause (1 cycle = 21 days) for a maximum of 6 cycles or until disease progression or unacceptable toxicity develops.

Facilities to treat anaphylaxis MUST be present when systemic anti-cancer therapy (SACT) is administered.

Day	Drug	Dose	Route	Diluent & Rate	Cycle
1 and 8	Gemcitabine	1250mg/m^2	IV infusion	250mL NaCl 0.9% over 30 minutes	Every 21 days

Note: Administration volumes and fluids have been standardised to facilitate electronic prescribing system builds.

ELIGIBILITY:

- Indications as above
- ECOG 0-2

EXCLUSIONS:

- Hypersensitivity to gemcitabine or any of the excipients
- Pregnancy
- Breast feeding

PRESCRIPTIVE AUTHORITY:

The treatment plan must be initiated by a Consultant Medical Oncologist

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TESTS:

Baseline tests:

- FBC, renal and liver profile

Regular tests:

- FBC prior to each treatment
- Renal and liver profile prior to each cycle

Disease monitoring:
Disease monitoring should be in line with the patient's treatment plan and any other test/s as directed by the supervising Consultant.

DOSE MODIFICATIONS:

- Any dose modification should be discussed with a Consultant.

Haematological:
Prior to commencing a new treatment cycle (i.e. day 1), ANC must be ≥1.0 x 10^9/L and platelets ≥100 x 10^9/L.

Table 1: Dose modifications for gemcitabine within a cycle (i.e day 8)

ANC (x 10^9 /L)		Platelet count (x 10^9 /L)		Other toxicity	Recommended dose of Gemcitabine
≥1	and	≥ 100			100 %
0.5- 1	or	50-100			75%
< 0.5	or	<50			Omit. Do not restart treatment until ANC ≥ 0.5 and platelets ≥ 50
ANC < 0.5 for ≥ 5 days **or** ANC < 0.1 for ≥3 days **or** Any incidence of febrile neutropenia	**or**	< 25	**or**	cycle delay of >1 week due to any toxicity	Reduce dose to 75% of the original cycle initiation dose for all subsequent cycles.

Renal and Hepatic Impairment:

Table 2: Dose modification of Gemcitabine in renal and hepatic impairment

Drug	Renal Impairment		Hepatic Impairment
Gemcitabine	**CrCl (mL/min)**	**Dose**	If bilirubin > 27 micromol/L, either start at 80% of the original dose and increase the dose if tolerated or start with full dose with active monitoring
	≥30	No dose adjustment is needed	
	<30	No need for dose adjustment is expected	
Renal and hepatic dose modifications as per Giraud et al 2023			

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Management of adverse events:

Table 3: Dose Modification of gemcitabine for Adverse Events

Adverse reactions	Recommended dose modification
Grade ≥ 2 Pneumonitis	Discontinue gemcitabine
Grade > 3 Non-haematological toxicity (except nausea/vomiting)	Therapy with gemcitabine should be withheld (until toxicity has resolved to grade ≤ 1) and may be resumed with 50% dose reduction or treatment discontinued at discretion of prescribing consultant.
Severe cutaneous adverse reactions (SCARs) e.g. Stevens-Johnson syndrome (SJS),toxic epidermal necrolysis (TEN), drug reaction with eosinophilia and systemic symptoms (DRESS) and acute generalised exanthematous pustulosis (AGEP)	Discontinue treatment

SUPPORTIVE CARE:

EMETOGENIC POTENTIAL:

As outlined in NCCP Classification Document for Systemic Anti Cancer Therapy (SACT) Induced Nausea and Vomiting - Available on the NCCP website

Gemcitabine: Low **(Refer to local policy).**

For information:
Within NCIS regimens, antiemetics have been standardised by Medical Oncologists and Haemato-oncologists and information is available in the following documents:

- NCCP Supportive Care Antiemetic Medicines for **Inclusion in NCIS** (Medical Oncology) - Available on the NCCP website
- NCCP Supportive Care Antiemetic Medicines for **Inclusion in NCIS** (Haemato-oncology) - Available on the NCCP website

PREMEDICATIONS: None usually required

OTHER SUPPORTIVE CARE: No specific recommendations

ADVERSE EFFECTS:

- Please refer to the relevant Summary of Product Characteristics (SmPC) for details.

DRUG INTERACTIONS:

- Current SmPC and drug interaction databases should be consulted for information.

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REFERENCES:

1. Foo, KF, Tan EH, Leong SS, et al. Gemcitabine in metastatic nasopharyngeal carcinoma of the undifferentiated type. Ann Oncol 2002;13:150-156.
2. EVIQ Protocol Summary for Nasopharyngeal recurrent or metastatic Gemcitabine- Using Protocol code 1510 revised October 2016.
3. BCCA Protocol Summary for Treatment of Loco-regionally Recurrent/Metastatic Nasopharyngeal Cancer not Amenable for Local Curative Therapy with Gemcitabine. Using Gemcitabine Protocol code HNNAVGEM revised November 2012
4. Giraud E L, Lijster B D, et al. Dose recommendations for anticancer drugs in patients with renal or hepatic impairment: an update. Available at: https://pubmed.ncbi.nlm.nih.gov/37269847/
5. NCCP Classification Document for Systemic Anti-Cancer Therapy (SACT) Induced Nausea and Vomiting. V6 2025. Available at: https://www.hse.ie/eng/services/list/5/cancer/profinfo/chemoprotocols/nccp-classification-document-for-systemic-anti-cancer-therapy-sact-induced-nausea-and-vomiting.pdf
6. Gemcitabine 1g Powder for Solution for Infusion Summary of Product Characteristics. Last updated 23/12/2024. Accessed September 2025. Available at: https://assets.hpra.ie/products/Human/27516/Licence_PA2315-092-002_23012024174158.pdf

Version	Date	Amendment	Approved By
1	07/11/2018		Prof Maccon Keane
2	23/10/2020	Reviewed	Prof Maccon Keane
3	02/12/2025	Regimen reviewed.Updated renal and hepatic dose modifications table to align with Giraud et al 2023. Update to Table 3 adverse events. Regimen update in line with NCCP standardisation.	Dr Colm Mac Eochagain

Comments and feedback welcome at oncologydrugs@cancercontrol.ie.

[i] This indication is outside the licensed indications for Gemcitabine in Ireland. Patients should be informed of the unlicensed nature of this indication and consented to treatment in line with the hospital's policy on the use of unlicensed medication and unlicensed or "off label" indications. Prescribers should be aware of their responsibility in communicating any relevant information to the patient and also in ensuring that the unlicensed or "off label" indication has been acknowledged by the hospital's Drugs and Therapeutics Committee, or equivalent, in line with hospital policy.

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